

WEEK 9:

➔ Thus, program is executed successfully...

Solved at: Wednesday, 20 May 2020, 8:57 AM
Status: Finished
Completion time: Wednesday, 20 May 2020, 9:07 AM
Time taken: 4 mins 12 secs
Name: SAKSHI WEL C.T.2025@BMS

Comments:

- Comment 1
- Answer
- Marked out of 1.00
- By Prof. Anurag

Write a program to convert a given Infix Expression into its equivalent Prefix Expression using a Stack.

Operators to be Supported:

- (Addition)
- (Subtraction)
- (Multiplication)
- (Division)
- (Exponentiation)

Operator Precedence & Associativity:

Operator Precedence	Associativity
^	Highest Right to Left
* /	Medium Left to Right
+ -	Lowest Left to Right

Sample Input:
 ab*c+d*e*f-g
 Sample Output:
 abc*d+efg*-

For example:

Input	Result
ab*c+d*e*f-g	abc*d+efg*-

Answer: (initially response: 0 %) [View answer](#)

```

1 #include <iostream.h>
2 #include <string.h>
3 #define MAX 20
4 int Stack[MAX], top = -1;
5 char exp[Max]; prec[Max];
6
7 void push(char sgn);
8 char pop();
9 int priority(char sgn);
10
11 void main()
12 {
13     int i;
14     printf("ab*c+d*e*f-g\n");
15     scanf("%s", exp);
16     for(i = 0; i < strlen(exp); i++)
17     {
18         if(exp[i] == '*' || exp[i] == '/')
19             printf("%c ", exp[i]);
20     }
21 }
```

Input	Expected	Got
✓ ab*c+d*e*f-g	abc*d+efg*-	✓

Result of test: ✓

➔ Thus, program is executed successfully...

Started on: Wednesday, 20 May 2020, 9:12 AM
By: Prithvi
Completed on: Wednesday, 20 May 2020, 9:20 AM
Time taken: 8 mins 28 secs
Score: 100/100 (100%)

1
 Correct
 Feedback
 Report a problem
 0/0
 0/0

Write a program to **evaluate a given Prefix Expression** using a **Stack**.
 The expression contains operators represented by variables, and their corresponding values will be provided separately.

Input Format:
 1. A string representing the **prefix expression**.
 2. The integer values of the operators in the **order of their appearance** in the expression.

Output Format:
 Print the **final evaluated result** of the prefix expression.

Sample Input:
 abc+*45
 2
 3
 4
 5

Sample Output:
 70

For example:

Input	Result
abc+*45	70
2	
3	
4	
5	

Answer: (Correctly explains 0/0%)

```

1 #include <iostream>
2 #include <string.h>
3 #include <stack>
4 using namespace std;
5 int Stack[100], top = -1;
6 char exp[100];
7 int StackLen = 0;
8 int pop();
9 int push();
10 int main()
11 {
12     int a, b, c, n;
13     printf("Enter n: "); scanf("%d", &n);
14     for (int i = 0; i < n; i++)
15     {
16         if (exp[i] == '+' || exp[i] == '-' || exp[i] == '*' || exp[i] == '/')
17         {
18             k = pop();
19             a = pop();
20             push(exp[i]);
21         }
22     }
  
```

Input	Expected	Got
abc+*45	70	✓
2		
3		
4		
5		

Passed all tests! ✓

➔ Thus, program is executed successfully...